

# Two Small-Order Classification Theorems for Signed Difference Sets

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## Abstract

A signed difference set assigns coefficients in  $\{0, 1, -1\}$  to a finite group so that every nonidentity periodic autocorrelation has the same value. We settle two clusters of cases that were open in the April 24, 2026 snapshot of the La Jolla Signed Difference Set Repository. First, a signed  $(32, 20, 4)$  difference set exists in an abelian group of order 32 if and only if the group is noncyclic. The six positive cases are given by explicit coefficient vectors, while the cyclic case is excluded by a complete  $C_8 \rightarrow C_{16} \rightarrow C_{32}$  quotient refinement. Second, no abelian group of order 36 admits a signed  $(36, 29, 4)$  difference set. This is proved by three short exhaustive arguments: an empty  $C_{18}$  quotient system excludes  $C_{36}$  and  $C_2 \times C_{18}$  simultaneously; an empty  $C_2 \times C_3^2$  quotient system excludes  $C_3 \times C_{12}$ ; and a solver-free two-marginal enumeration excludes  $C_6 \times C_6$ . All searches use exact arithmetic, expose their complete finite search spaces, and are accompanied by regenerating verification programs. An external computational review reproduced the witnesses and the finite computations supporting these two theorems; its scope is stated separately below.

**Keywords.** Signed difference set; finite abelian group; quotient enumeration; autocorrelation; computer-assisted proof.

**2020 Mathematics Subject Classification.** 05B10; 05B20; 20K01.

## 1 Introduction

Let  $G$  be a finite group of order  $v$ . A signed subset of  $G$  is a group ring element

$$D = \sum_{g \in G} a_g g, \quad a_g \in \{0, 1, -1\}.$$

Following Gordon [1],  $D$  is a signed  $(v, k, \lambda)$  difference set when exactly  $k$  coefficients are nonzero and

$$DD^{(-1)} = (k - \lambda)e + \lambda G. \tag{1}$$

Equivalently, its periodic autocorrelation is  $k$  at the identity and  $\lambda$  at every nonidentity shift. Signed difference sets contain ordinary difference sets and circulant weighing matrices as special cases; further constructions were obtained by He, Chen, and Ge [2].

The starting point of this work was the La Jolla Signed Difference Set Repository [3], frozen at commit `e3bf810c5ee6826cf5030f983f6adf23b0ffd20e` on April 24, 2026. The following two theorems concern ten entries marked `Open` in that snapshot.

**Theorem 1** (Order 32 classification). *Let  $G$  be an abelian group of order 32. A signed  $(32, 20, 4)$  difference set exists in  $G$  if and only if  $G$  is noncyclic. The positive direction is certified by six explicit constructions; the negative direction is certified by a complete quotient refinement for  $C_{32}$ .*

**Theorem 2** (Order 36 classification). *No abelian group of order 36 admits a signed  $(36, 29, 4)$  difference set.*

The results are existence classifications, not classifications up to equivalence. In particular, the six positive files at order 32 contain one representative per group; we do not claim that they contain all orbits under translations, automorphisms, and global sign.

The first version of this note used the repository's cyclic  $C_{36}$  nonexistence and a proof-producing SAT calculation for the final  $C_6 \times C_6$  case. Independent review by Arévalo [4] produced two simplifications now incorporated here: a one-step  $C_{18}$  quotient exclusion for the cyclic case and a direct solver-free exhaustion for  $C_6 \times C_6$ . The original DRAT certificate is retained as an independent supplementary check, but neither Theorem 2 nor its presentation now depends on SAT.

## 2 Arithmetic constraints and quotient projection

For  $h \in G$ , write

$$\text{corr}_D(h) = \sum_{g \in G} a_g a_{h^{-1}g}.$$

Equation (1) is equivalent to  $\text{corr}_D(e) = k$  and  $\text{corr}_D(h) = \lambda$  for  $h \neq e$ . Applying the trivial character gives

$$\left( \sum_{g \in G} a_g \right)^2 = k + (v - 1)\lambda. \quad (2)$$

For every nonprincipal character  $\chi$  of an abelian group,

$$\left| \sum_{g \in G} a_g \chi(g) \right|^2 = k - \lambda. \quad (3)$$

Thus a global sign may normalize the coefficient sum to be nonnegative. For  $(32, 20, 4)$  the sum is 12, so a solution has 16 positive and 4 negative coefficients. For  $(36, 29, 4)$  the sum is 13, so a solution has 21 positive and 8 negative coefficients.

The computational reductions use the following elementary projection. Its sum and squared-norm consequences are also instances of the intersection-number relations in Lemma 5.2 of Gordon [1].

**Lemma 1** (Quotient projection). *Let  $K \trianglelefteq G$  have order  $m$ , let  $Q = G/K$ , and define the cell-sum projection*

$$B = \sum_{q \in Q} b_q q, \quad b_q = \sum_{g \in q} a_g.$$

*If  $D$  is a signed  $(v, k, \lambda)$  difference set, then*

$$BB^{(-1)} = (k - \lambda)e_Q + m\lambda Q. \quad (4)$$

*Consequently,*

$$\sum_q b_q = \sum_g a_g, \quad \sum_q b_q^2 = k + (m - 1)\lambda,$$

*and every nonzero quotient shift has correlation  $m\lambda$ . Each cell is an integer in  $[-m, m]$ .*

*Proof.* The natural map  $\mathbb{Z}[G] \rightarrow \mathbb{Z}[Q]$  is a ring homomorphism. Applying it to Equation (1) sends  $e$  to  $e_Q$  and sends the formal sum  $G$  to  $mQ$ , which gives Equation (4). Reading the identity coefficient gives the sum of squares, and applying the trivial character gives the total sum. The cell bound follows because each fiber contains  $m$  coefficients in  $\{0, 1, -1\}$ .  $\square$

### 3 What the exhaustive programs enumerate

We describe the finite searches explicitly because a failed heuristic search would not prove nonexistence. For a quotient  $Q$  of size  $r$  and kernel size  $m$ , the basic enumerator performs the following steps.

- E1.** Generate every vector  $b \in [-m, m]^r \cap \mathbb{Z}^r$  having the required total sum and squared norm from Lemma 1. A depth-first traversal uses necessary feasibility bounds: the remaining cells must be able to reach the remaining sum and squared norm, and Cauchy–Schwarz requires  $(\text{remaining sum})^2 \leq (\text{remaining cells})(\text{remaining norm})$ .
- E2.** For each terminal vector, calculate all  $r$  quotient correlations directly and retain it precisely when the identity and every nonidentity value agree with Equation (4).
- E3.** When a retained cell of size  $2m$  is refined into two cells of size  $m$ , generate every bounded pair whose sum is the parent value. Repeat until the final entries lie in  $\{0, 1, -1\}$ , then check the support size and every full-group correlation.

Every loop is finite and uses exact integer arithmetic. Translation and automorphism orbits are used only where their actions are proved to lift to the next quotient. Regenerating checkers reconstruct the search spaces rather than trusting saved candidate counts. Positive witnesses are checked by two structurally separate implementations: one uses tuple-valued group elements and a dictionary operation, while the other uses mixed-radix integer indices.

### 4 The six constructions at order 32

There are seven abelian groups of order 32. The six noncyclic groups all admit signed  $(32, 20, 4)$  difference sets. Table 1 points to one full coefficient vector for each group. Every vector contains 16 entries  $+1$ , four entries  $-1$ , and twelve zeros, and both validators return the correlation vector  $(20, 4, \dots, 4)$ .

Table 1: Explicit order-32 witnesses in the accompanying data.

| Group               | Witness file under <code>artifacts/witnesses/</code> |
|---------------------|------------------------------------------------------|
| $C_2 \times C_{16}$ | <code>sds_32_20_4_2_16.json</code>                   |
| $C_4 \times C_8$    | <code>sds_32_20_4_4_8.json</code>                    |
| $C_2^2 \times C_8$  | <code>sds_32_20_4_2_2_8.json</code>                  |
| $C_2 \times C_4^2$  | <code>sds_32_20_4_2_4_4.json</code>                  |
| $C_2^3 \times C_4$  | <code>sds_32_20_4_2_2_2_4.json</code>                |
| $C_2^5$             | <code>sds_32_20_4_2_2_2_2_2.json</code>              |

The JSON files record the invariant factors, coordinate convention, full coefficient vector, positive and negative point sets, and complete autocorrelation output. They are also exported as  $[P, N]$  coordinate lists in the format of the La Jolla repository. Appendix A records the six constructions explicitly. Since existence is verified by substitution in Equation (1), the six files prove the positive direction of Theorem 1.

### 5 Excluding the cyclic group of order 32

Let  $D$  be a hypothetical signed  $(32, 20, 4)$  difference set in  $C_{32}$ . Projection first to  $C_8$  uses a kernel of order 4. The cells therefore lie in  $[-4, 4]$ , have sum 12, squared norm  $20 + 3 \cdot 4 = 32$ , and nonzero correlation 16. The complete enumeration from Section 3 has the counts in Table 2.

Table 2: Complete cyclic quotient refinement for  $(32, 20, 4)$ .

| Stage                                            | Examined | Surviving |
|--------------------------------------------------|----------|-----------|
| $C_8$ : bounded vectors after sum and norm       | 9,528    | 56        |
| Affine orbits on $C_8$                           | 56       | 5 orbits  |
| $C_{16}$ refinements of the five representatives | complete | 12        |
| $C_{32}$ coefficient refinements                 | 248,832  | 0         |

The five  $C_8$  orbit sizes are 8, 16, 8, 16, 8. The affine action is generated by translations and multiplication by units. Both operations lift through  $C_{16}$  to  $C_{32}$  because reduction  $(\mathbb{Z}/32\mathbb{Z})^\times \rightarrow (\mathbb{Z}/8\mathbb{Z})^\times$  is onto. For each representative, every bounded split into two  $C_{16}$  cells is generated and checked. All twelve  $C_{16}$  survivors in the normalized parent slices are retained; no second symmetry quotient is imposed. Each is then split in every possible way into two coefficients in  $\{0, 1, -1\}$ , with exact support checked before all 31 full correlations. No full vector survives.

As an independent robustness check, Arévalo refined all 56  $C_8$  solutions without any orbit reduction. That calculation produced 144  $C_{16}$  survivors and tested 2,985,984 final refinements, again with no solution [4]. Thus the negative direction does not depend in practice on the affine reduction. This completes the proof of Theorem 1.

## 6 The order 36 classification

The abelian groups of order 36 are

$$C_{36}, \quad C_2 \times C_{18}, \quad C_3 \times C_{12}, \quad C_6 \times C_6.$$

For  $(36, 29, 4)$ , Equation (2) gives normalized sum 13. We exclude the four groups in three computations.

### 6.1 One $C_{18}$ system excludes two groups

Both  $C_{36}$  and  $C_2 \times C_{18}$  have a quotient  $C_{18}$  with kernel of order 2. A projected vector must therefore lie in  $[-2, 2]^{18}$ , have sum 13, squared norm  $29 + 4 = 33$ , and correlation 8 at every nonzero shift.

Rather than enumerate  $5^{18}$  vectors, the verifier constructs the complete  $C_9$  and  $C_2$  marginal systems first. The  $C_9$  system has cell bound 4, squared norm 41, and nonzero correlation 16. Among 106,353 vectors satisfying sum and norm, its only nine solutions are the nine translations of

$$(-3, 2, 2, 2, 2, 2, 2, 2, 2).$$

The  $C_2$  system has precisely the two solutions  $(4, 9)$  and  $(9, 4)$ . For every one of the  $9 \cdot 2$  marginal pairs, the program enumerates every split

$$b_i + b_{i+9} = t_i, \quad b_i, b_{i+9} \in [-2, 2],$$

and retains the required even and odd sums. There are 19,152 marginal-compatible refinements, of which 7,560 have squared norm 33. Direct evaluation of all 18 cyclic correlations leaves no solution. Consequently neither  $C_{36}$  nor  $C_2 \times C_{18}$  admits the required signed difference set.

### 6.2 The group $C_3 \times C_{12}$

Project  $C_3 \times C_{12}$  onto  $C_2 \times C_3^2$  with kernel order 2. The real marginal is one of  $(4, 9)$  and  $(9, 4)$ , and the  $C_3^2$  marginal is a translation of  $(-3, 2, \dots, 2)$ . Translation in the two quotient factors

acts independently and lifts to  $C_3 \times C_{12}$ , so one pair suffices. The local combined-quotient checker uses this normalization; Arévalo’s external review additionally checked every marginal pair without it.

For the normalized pair, each of the nine  $C_3^2$  cells is split into two entries in  $[-2, 2]$ , while the row sums are fixed to  $(4, 9)$ . Exactly 420 vectors satisfy these marginals and squared norm 33. None has correlation 8 at all 17 nonzero shifts of  $C_2 \times C_3^2$ . The quotient system is empty, so no full signed difference set exists in  $C_3 \times C_{12}$ .

### 6.3 A direct solver-free search for $C_6 \times C_6$

Use the Chinese remainder isomorphism

$$C_6 \times C_6 \cong C_2^2 \times C_3^2.$$

The complete  $C_3^2$  quotient system consists of the nine translations of  $(-3, 2, \dots, 2)$ . The complete  $C_2^2$  quotient system consists of the four translations of  $(7, 2, 2, 2)$ . The direct search examines all  $9 \cdot 4 = 36$  pairs, so it uses no symmetry normalization.

Fix a marginal pair  $(T, R)$ . For each of the nine  $C_3^2$  positions  $j$ , generate every pattern  $p_j \in \{-1, 0, 1\}^4$  satisfying  $\sum_z p_j(z) = T_j$ . Split the nine fibers into groups of four and five. For each half, store the four partial  $C_2^2$  sums; join two half-vectors exactly when their partial sums add to  $R$ . This meet-in-the-middle step is only an indexing optimization: it generates every full vector with the chosen marginals exactly once.

Across all 36 marginal pairs, the joins produce 16,964,640 full vectors. Each is filtered by the 35 nonidentity correlation equations in exact integer arithmetic. The dot products have magnitude at most 36, so the signed 64-bit batch operations cannot overflow. Every candidate has sum 13; if all 35 nonidentity correlations were 4, their sum would force the identity correlation to be  $13^2 - 35 \cdot 4 = 29$ . Thus no separate support filter is omitted. No vector survives. The original SAT encoding and its checked DRAT trace reach the same conclusion, but are not needed for this proof. The four abelian groups have now all been excluded, proving Theorem 2.

## 7 Independent replication and verification limits

Arévalo’s review [4] reimplemented the checks from the definition, independently of the original project’s two witness validators, quotient programs, and SAT encoder. The present  $C_6 \times C_6$  program was subsequently adapted from that review, so it is not a third independent implementation of the direct search. The review reports the following.

- All 16 witnesses from the broader census pass a third checker written against Gordon’s convention.
- Every count in Sections 5 and 6 was reproduced. For the cyclic order-32 and the order-36 quotient systems, all symmetry reductions were removed and the same nonexistence decisions were obtained.
- The direct  $C_6 \times C_6$  search over 16,964,640 candidates was the independent source of the simplified proof now used above.
- A cell-by-cell comparison of the larger 68-entry census found 58 agreements with Arévalo’s earlier census and no disagreement; the ten cases in Theorems 1 and 2 are exactly the ten his census had left open.

The review did not audit the CNF encoder or the contents of the large Zenodo archive. That limitation no longer bears on Theorem 2, because the direct search replaces the SAT argument. The review itself used substantial AI assistance, disclosed by its author; its evidentiary claims are backed by committed deterministic programs and outputs rather than model responses.

## 8 Exact computation and reproducibility

The accompanying source tree contains the manuscript, witnesses, exact quotient programs, and machine-readable outputs. Table 3 connects each part of the two theorems to the finite calculation that checks it. The mathematical reductions and completeness arguments are given in Sections 3–6; agreement with a stored count alone would not establish that a search covers every possible solution.

Table 3: Verification targets for the two classification theorems.

| Claim                                          | Exact check and expected result                                                                                                                             |
|------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Six positive order-32 cases                    | Both validators recompute support 20 and all 32 correlations of each printed construction.                                                                  |
| No solution in $C_{32}$                        | 9,528 bounded $C_8$ candidates yield 56 solutions in five affine orbits; 12 normalized $C_{16}$ refinements yield 248,832 final candidates and no solution. |
| No solution in $C_{36}$ or $C_2 \times C_{18}$ | All 18 marginal pairs yield 19,152 refinements; 7,560 have norm 33, and none satisfies the $C_{18}$ correlations.                                           |
| No solution in $C_3 \times C_{12}$             | The normalized quotient system has 420 marginal-and-norm candidates and no solution.                                                                        |
| No solution in $C_6 \times C_6$                | All 36 marginal pairs yield 16,964,640 full vectors and no solution.                                                                                        |

From the project root, the following commands rerun the witness audit and the four finite computations. The environment requires Python 3; the direct  $C_6 \times C_6$  calculation additionally uses NumPy 2.0.2.

```
.venv/bin/python scripts/audit_final_census.py
.venv/bin/python scripts/verify_v32_remaining_quotients.py
.venv/bin/python scripts/verify_v36_29_4_c18_quotient.py
.venv/bin/python scripts/verify_v36_29_4_combined_quotients.py
.venv/bin/python scripts/verify_v36_29_4_c6x6_direct.py
```

The first command checks the full census and both witness validators; it also requires the separate CNF/DRAT files when auditing their stored hashes. The remaining commands regenerate the cyclic order-32 ladder, the empty  $C_{18}$  system, the combined quotients (including  $C_3 \times C_{12}$ ), and the full  $C_6 \times C_6$  search, respectively. The lightweight package provides `verify_package.py` for checking its file manifest and all 16 witnesses without the large trace archive.

The calculations use integer arithmetic, with no floating-point tolerance, randomized acceptance test, or language-model judgment. Python integer operations are unbounded. The only fixed-width correlation batches use signed 64-bit integers on 36 coefficients in  $\{-1, 0, 1\}$ ; each product has magnitude at most one and each sum at most 36. Early rejection uses necessary equations, so a rejected candidate cannot be a solution. The saved outputs document the runs, while the source describes how to recreate their finite search spaces.

These computational checks do not constitute a formal verification of the Python interpreter, NumPy, or the mathematical reductions. Repeating an AI-assisted calculation is also distinct from an independent human proof audit. The external review and the independently implemented witness validators provide the additional checks described above.

### 8.1 Relation to the larger census and archived releases

The two theorems emerged from a census of all 68 entries with group order at most 36 marked **Open** in the frozen repository snapshot. That broader project records 16 existence results and 52 nonexistence results. The ten cases treated here are the principal novelty-supported mathematical

contribution; the other 58 are independent replications of an exact public predecessor. Literature and repository searches found no earlier resolution of the ten cases, but absence from a search is not a mathematical proof of novelty.

The public version 1.0 repository is available at [5]. The original 1.1 GB collection of CNF formulas and DRAT traces is archived at Zenodo [6]. It remains useful as an independent certificate set for the larger census, even though the order-36 theorem now has a solver-free proof. The source package accompanying this submission contains the additional order-36 checks described here; the public version 1.0 release and its DOI should not be mistaken for an archive of this revised manuscript.

## 9 Use of generative AI and author responsibility

OpenAI Codex was used extensively throughout this project, including to propose candidate character, quotient, and SAT search strategies; write and revise search programs, tests, and validators; execute local experiments; organize evidence and reproducibility materials; and draft and revise the mathematical exposition. Its contribution therefore extended to research exploration and implementation as well as language editing. The particular model version used for every historical step is not established by the project’s disclosure record; no single model version is asserted here.

The evidentiary basis is the explicit constructions, complete finite enumerations, and deterministic checks described in Section 8. An AI-generated argument, a failed search, or an unchecked solver answer is not accepted as evidence. The two theorems in this revision use the direct enumerations; the checked DRAT certificates support additional census entries and a supplementary check of the  $C_6 \times C_6$  case. Arévalo’s independently developed review also disclosed substantial AI assistance, with the specific coverage and limitations stated above.

The human author selected the objective, authorized the computational work, and reviewed the framing, and remains responsible for the claims and any submission or public release. This statement records the documented roles and checks; it does not assert a separate line-by-line human verification of every argument or program. Tan’s recent work on circulant weighing matrices [7] provides a related example of reporting exact computations alongside an explicit AI-use disclosure; the attribution here describes this project’s own workflow.

**Acknowledgments.** The author thanks Daniel M. Gordon for discussing the prior-art search and mathematical framing, clarifying the repository data format, and encouraging journal submission. The author thanks Fabian Arévalo for a detailed independent reproduction, for the  $C_{18}$  observation, and for the solver-free  $C_6 \times C_6$  enumeration used in the revised proof.

**Supplementary information.** Online Resource 1 is a reproducibility package containing the exact programs, witnesses, machine-readable outputs, run instructions, and an integrity manifest for the computations used in the two theorems.

## Statements and Declarations

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**Competing interests.** The author has no relevant financial or non-financial interests to disclose.

**Data availability.** The coefficient vectors, machine-readable outputs, and integrity manifests supporting the two classification theorems are included in the accompanying reproducibility package (Online Resource 1). The version 1.0 CNF and DRAT archive for the broader census is publicly available from Zenodo at <https://doi.org/10.5281/zenodo.21901581>. That archive is a historical release and is not presented as an archive of the present revision.

**Code availability.** The exact Python programs needed to validate the witnesses and regenerate the finite searches used in the two theorems are included in Online Resource 1. The public project repository is available at <https://github.com/NicolasMasselot/certified-small-sds-census>; the submission package records the revision-specific files separately from the public version 1.0 release.

**Author contribution.** Nicolas Masselot selected the research objective, directed the computational investigation, reviewed the mathematical framing, and takes responsibility for the manuscript and its submission. The substantial roles of generative AI in research exploration, implementation, experiment execution, evidence organization, and drafting are described in the preceding section. Fabian Arévalo’s separately authored review and the components adapted from it are attributed above and in the cited review artifact.

## A Explicit order-32 constructions

Coordinates are written in invariant-factor order. In each case  $D = \sum_{g \in P} g - \sum_{g \in N} g$ . Direct substitution gives support 20 and correlation 4 at every nonidentity shift.

$C_2 \times C_{16}$ .

$$\begin{aligned} P &= \{(0, 5), (0, 7), (0, 10), (0, 11), (0, 13), (0, 15), (1, 0), (1, 1), \\ &\quad (1, 3), (1, 4), (1, 7), (1, 8), (1, 9), (1, 10), (1, 12), (1, 15)\}, \\ N &= \{(0, 2), (0, 3), (1, 2), (1, 11)\}. \end{aligned}$$

$C_4 \times C_8$ .

$$\begin{aligned} P &= \{(0, 4), (0, 5), (0, 6), (1, 0), (1, 1), (1, 2), (1, 3), (1, 5), \\ &\quad (1, 6), (2, 1), (2, 2), (2, 4), (3, 2), (3, 4), (3, 6), (3, 7)\}, \\ N &= \{(0, 0), (2, 0), (3, 1), (3, 5)\}. \end{aligned}$$

$C_2^2 \times C_8$ .

$$\begin{aligned} P &= \{(0, 0, 1), (0, 0, 2), (0, 0, 3), (0, 0, 7), (0, 1, 0), (0, 1, 3), \\ &\quad (0, 1, 4), (0, 1, 6), (0, 1, 7), (1, 0, 4), (1, 0, 5), (1, 1, 1), \\ &\quad (1, 1, 3), (1, 1, 4), (1, 1, 6), (1, 1, 7)\}, \\ N &= \{(0, 1, 1), (1, 0, 0), (1, 0, 6), (1, 1, 0)\}. \end{aligned}$$

$C_2 \times C_4^2$ .

$$\begin{aligned} P &= \{(0, 0, 3), (0, 1, 0), (0, 1, 1), (0, 2, 0), (0, 2, 1), (0, 2, 2), \\ &\quad (0, 3, 1), (1, 0, 0), (1, 1, 0), (1, 1, 1), (1, 1, 3), (1, 2, 0), \\ &\quad (1, 2, 3), (1, 3, 1), (1, 3, 2), (1, 3, 3)\}, \\ N &= \{(0, 0, 0), (0, 0, 2), (0, 1, 2), (1, 2, 1)\}. \end{aligned}$$

$C_2^3 \times C_4$ .

$$\begin{aligned} P = \{ & (0, 0, 0, 0), (0, 0, 0, 2), (0, 0, 0, 3), (0, 0, 1, 1), \\ & (0, 0, 1, 2), (0, 0, 1, 3), (0, 1, 0, 0), (0, 1, 0, 1), \\ & (0, 1, 1, 1), (0, 1, 1, 2), (1, 0, 0, 0), (1, 0, 0, 2), \\ & (1, 0, 1, 2), (1, 1, 0, 1), (1, 1, 0, 2), (1, 1, 0, 3) \}, \\ N = \{ & (0, 1, 0, 3), (0, 1, 1, 0), (1, 0, 0, 1), (1, 1, 1, 1) \}. \end{aligned}$$

$C_2^5$ .

$$\begin{aligned} P = \{ & (0, 0, 1, 1, 0), (0, 0, 1, 1, 1), (0, 1, 0, 0, 0), (0, 1, 0, 0, 1), \\ & (0, 1, 0, 1, 1), (0, 1, 1, 0, 1), (1, 0, 0, 0, 0), (1, 0, 0, 1, 0), \\ & (1, 0, 0, 1, 1), (1, 0, 1, 0, 0), (1, 0, 1, 1, 1), (1, 1, 0, 0, 1), \\ & (1, 1, 0, 1, 0), (1, 1, 0, 1, 1), (1, 1, 1, 0, 0), (1, 1, 1, 1, 0) \}, \\ N = \{ & (0, 0, 0, 1, 1), (0, 0, 1, 0, 1), (1, 0, 1, 0, 1), (1, 1, 1, 0, 1) \}. \end{aligned}$$

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